A robot can operate in one of n modes. You are given a list of m commands that change the robot from one mode to another; the reverse command may not exist. The manufacturer claims: "No matter which commands you execute, you can always restore the robot to the mode it was in before you started." Give an efficient algorithm to check this claim. (Note: there is an example input at the bottom of the page, but you should answer these questions for a general input.)

- What are the vertices? What does each vertex represent?

- What are the edges? Are they directed or undirected?

- If the vertices and/or edges have associated values, what are they?

- **What is the precise relationship between your graph and the stated problem?**

- What standard graph algorithm are you using to solve the problem?

- What is the running time of your entire algorithm, *including* the time to build the graph, *as a function of the original input parameters*?

Here is an example you may want to explore:

Command	Starting mode	Resulting mode
Begin charging	Standby	Charging
Stop charging	Charging	Standby
Switch to mop	Vacuuming	Mopping
Begin drying	Mopping	Drying
Return to base	Drying	Returning
Resume cleaning	Returning	Vacuuming